

Is acoustic re-identification recognising the animal, or the recording?

Four diagnostics applied to 13 public datasets from nine species and seven studies, with BirdNET and 35 other neural networks

As commonly tested Under the diagnostic

THE PROBLEM

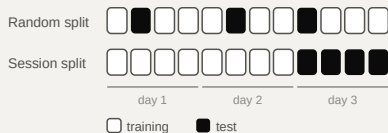
Two common practices can inflate reported re-ID accuracy

a. Each animal is recorded at its own site



Identity and recording site are linked, so a classifier can score highly by recognising the place.

b. Clips are split at random

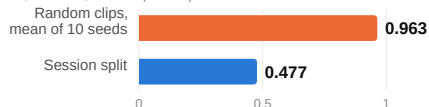


Training and test clips share sessions, so the score no longer measures recognising an animal in a new session.

1 Split by recording session

Does accuracy hold in a new recording session?

Little owl, BirdNET, same clips and per-bird counts



Mixing clips from the same sessions into training and test raised accuracy by 0.20 to 0.51 in 18 comparisons with matched clips, counts and models (three species).

3 Hold the recording fixed

Where animals share a microphone, does re-ID survive?

- ✓ Zebra finches, group of 4 identity signal beyond recording
- ? Zebra finches, group of 8 recording not ruled out
- ? Rooks, one aviary recording not ruled out

Only one dataset gave clear evidence beyond recording. Both finch groups lack cross-recording same-bird pairs within seven days; changes over time remain a confound.

2 Take the animal away

Can ambient sound alone identify individuals?

Chiffchaff within one year, BirdNET, balanced accuracy



Background alone identified the birds for 35 of 38 embeddings and controls.

4 Let strangers in

Are known animals still named when others call?

Little owl · Perch 2.0 · nearest clip · four enrolled birds



Highest median in wild session-split tests: 46%, across 36 networks and two scorers; 16 role allocations. Known calls accepted and correctly named; stranger rates can exceed 10%.

Accuracy reported without these four diagnostics can substantially overstate acoustic re-ID.

No new models are needed; each diagnostic requires suitable data. Test before planning population monitoring.